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Incentive and Disincentive Experimentation for Income Maintenance Policy Purposes: Note

By EDGAR K. BROWNING*

In a recent article in this *Review*, Guy H. Orcutt and Alice G. Orcutt discuss the use of experimental techniques in predicting the quantitative outcomes of income maintenance policies. The experimental procedure recommended is to take a random sample from the entire group which would be affected by the policy under consideration. The sample is to be subdivided into two sub-samples, the first to receive negative income tax transfers and the second to act as a control group and receive no such transfers. Orcutt and Orcutt assert that the results of such a study provide the basis for an unbiased estimate of the magnitude of the disincentive effects of the policy under consideration:

From the first sample we could obtain an unbiased estimate of what the earned income of the entire group would have been if it had been exposed to a negative income tax. From the second sample we could get an unbiased estimate of what the earned income of the same group for the same time period would have been if the entire group had not been exposed to the negative income tax.
[p. 758]

The purpose of this note is to show that this assertion is incorrect—that the results of such an experiment cannot be generalized to provide reliable information concerning the disincentive effects of an income maintenance policy. The reason is that the aggregate market effects of such a policy will not occur in the sample study, and cannot be inferred from the experimental results. To illustrate the argument, I will assume that a specific negative income tax (*NIT*) plan is the policy under consideration. In addition,

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it will be assumed that there are no other welfare programs in existence, so that additional problems relating to the substitution of a *NIT* for other programs will not be encountered.

In Section I, I analyze briefly the effects of the *NIT* policy implemented on a national basis in a competitive economy where individuals are free to vary their work effort at the market wage rate. These effects are contrasted with the predictions of an experimental study of the type outlined above. In Section II, I allow for institutional rigidities in labor markets, and show how these will affect the outcomes of a national *NIT* policy and of an experimental study.

I. Aggregate Market Effects

Let *ON* in Figure 1 be the budget constraint relating income to hours worked for a household. The *NIT* shifts the constraint to *MBN* where *OM* is the income guarantee and *B* is at the breakeven level of income. The initial effect of this is that the household moves from *A* to *C*, reducing work effort from *H*₁ to *H*₃. Both the income effect

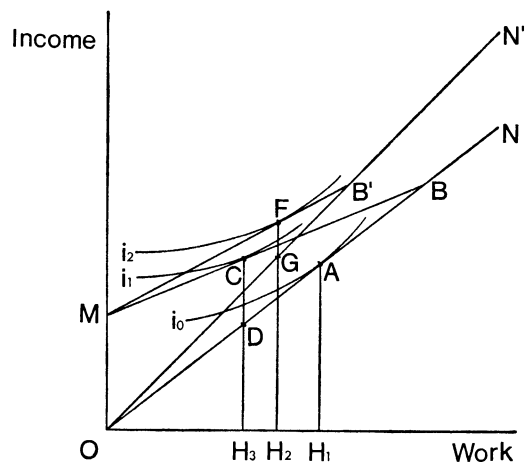


FIGURE 1

(following a higher income at each level of work effort over the relevant range and assuming leisure is not inferior) and the substitution effect (resulting from a lower net wage rate) encourage less work. All households with an income less than the break-even level will have a similar incentive to reduce their work effort.¹

This initial reduction in work effort on the part of the transfer recipients will show itself in labor markets as a reduction in the supply of labor. In Figure 2, the DD and SS curves represent the aggregate demand and supply curves for the type of labor which will receive NIT transfers. Prior to implementation of the NIT , W_1 is the market wage rate (which can best be thought of as a proxy for the level of wage rates generally), L_1 is the quantity of labor supplied and demanded, and OW_1KL_1 is the earned income of these workers. The NIT produces a shift in the supply curve to $S'S'$. The market wage rises to W_2 , the equilibrium quantity of labor supplied falls to L_2 , and earned income becomes OW_2JL_2 . Earned income may be greater or less than formerly, depending on whether the elasticity of demand is smaller or larger than unity.

The impact of this change in the market wage rate on the individual household's budget constraint is illustrated in Figure 1. The straight line ON' becomes the new constraint relating earned income and hours of work, and $MB'N'$ is the new budget constraint including the NIT transfer. (Note that B' occurs at the same level of earned income as B : the break-even level, which is defined as the level of earned income at which no transfer is received, has not been changed.) Suppose that the household moves from C to F , increasing work effort from H_3 to H_2 and earned income from DH_3 to GH_2 .² Work effort will, of course, be reduced below

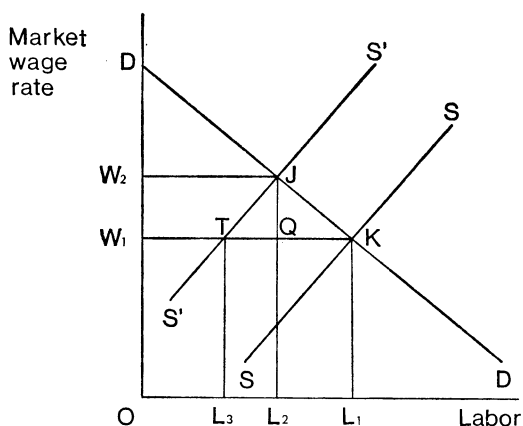


FIGURE 2

H_3 in response to the higher market wage rate if the supply curve is backward bending. Figure 1 is drawn with an increase to H_2 to conform to the upward sloping supply curve in Figure 2.

This, in perhaps overly concise form, is the outcome that would be predicted from standard economic theory. Now let us consider the predictions which an experimental study of this NIT plan would generate. From among those initially supplying the L_1 units of labor, a random sample is drawn and subjected to the payment of NIT transfers. Suppose that our representative household in Figure 1 is selected for this treatment. Then its budget constraint shifts to MBN , and it moves to point C .

And this is all that happens. For the sample will always be a negligible part of the entire market, so that there will be no noticeable pressure on the market wage rate to change.

Given that the sample is random, and that there are no "Hawthorne"-type problems,³ the experiment would predict a reduction in the quantity of labor supplied from L_1 to L_3 in Figure 2. This corresponds to the reduction from H_1 to H_3 in Figure 1. What would be estimated, in fact, is the amount by which the quantity of labor supplied would fall at the existing market wage rate, i.e., the horizontal shift in the supply curve. It would

¹ It is often the case that households with incomes above the break-even level will have an incentive to reduce work effort also, but this point will be ignored.

² The shift in the supply curve can be illustrated in Figure 1 by drawing in the two price consumption curves. The price consumption curve when the NIT is in effect lies to the left (at lower work effort) of the original curve at each market wage rate, at least until the household earns an income above the break-even level.

³ For an explanation and discussion of the "Hawthorne Effect," see Orcutt and Orcutt, pp. 759-61.

also be predicted that earned income would fall to OW_1TL_3 .

Now we can easily contrast the experimental predictions with the actual outcome. Referring to Figure 2 again, the prediction regarding work effort is a fall to L_3 . In actuality, work effort may be greater or less than L_3 , following enactment of a national *NIT* program. It will be greater if the supply curve is upward sloping through T (as it is drawn) and it will be less if the supply curve is backward bending.⁴ The prediction regarding earned income is a fall to OW_1TL_3 . In fact, earned income may not fall from its initial level at all: if the demand curve is inelastic, earned income will rise following enactment of the *NIT* on a national basis, as noted earlier. In any event, earned income will always be greater than OW_1TL_3 ,⁵ since the change in the market wage rate produces a movement up the new supply curve from point T , and earned income always increases with a move up a supply curve, whatever its slope.

In addition, the experiment would provide a prediction of the fall in national product of TKL_1L_3 , the estimated reduction in labor supply weighted by the initial market wage rate. The actual reduction in Figure 2 is JKL_1L_2 : whether this is greater or less than the predicted reduction depends on (in general) the elasticities of both the demand and the supply curves.

A final point might be noted. A *NIT* will generally change the rate of return for other factors of production (which include more highly paid labor which does not receive *NIT* transfer payments). If we assume that the other factors are in completely inelastic supply,⁶ then the income of these factors will fall from DW_1K to DW_2J , or by W_1W_2

JK .^{7,8} This is simply because the marginal productivity of these factors falls when the quantity of the factor they are combined with in production (labor receiving *NIT* transfers) is reduced. An experiment will provide no way to estimate this effect on the rate of return for other factors of production.

Thus, in this world where individuals are free to vary work effort, an experimental study will not provide unbiased estimates of the effects of a *NIT* policy. The reason is simply that the experiment will, at best, just estimate the horizontal shift in the supply curve, but will give no indication of the consequences of this shift—which is what we would really like to know.

It would obviously be helpful to know how much the supply curve will shift. It might be argued that if the shift is relatively small, then the aggregate market repercussions discussed here will be of little importance. In the next section, however, I will argue that it is doubtful that an experiment will provide even this piece of information.

II. Group Versus Individual Adjustment

In the last section it was assumed that individual households could vary freely the number of hours worked. However, in reality, individual households cannot, in general, work any number of hours they please. There are well-known institutional rigidities in labor markets which preclude this possibility. These rigidities usually take the form of all-or-nothing offers from employers. For example, an employer will offer a wage rate contingent on the employee working a forty-hour week. The employee must perform forty hours of work per week, or else he will not for long hold the job.

Taking account of such institutional rigidities does not alter the effects that would be expected from a national *NIT* program. Whereas no single individual may be able to influence (say) the length of the work week, the employers have incentive to tailor their

⁴ If the experiment is testing a number of *NIT* plans with different marginal tax rates, it may give some information about the elasticity of the supply curve. However, this possibility is weakened by the consideration raised in Section II.

⁵ If the experimental results are used to estimate the total government outlay on the *NIT*, then an underestimate of earned income means that the outlay will be overestimated since the transfers vary inversely with earned income under a *NIT* arrangement.

⁶ This has actually been implicit all along. If the quantity of other factors changes, then the demand curve in Figure 2 will shift.

⁷ This, of course, does not show the additional fall in income resulting from the higher taxes necessary to finance the program.

⁸ This does not mean that the rate of return for each and every factor will fall. Factors which are very good substitutes for the type of labor receiving *NIT* transfers may rise in price.

offers to the desires of the entire group of workers. Since the *NIT* will result in a general desire for shorter hours of work, the institutional rigidities will break down as employers compete for workers in markets where the aggregate supply conditions have changed. For this reason, institutional rigidities such as a fixed work week must be considered as variables in the analysis of a general change in the desires of workers.

These institutional rigidities, however, will not be affected by an experimental study. The households in the sample may be unable to adjust their work effort to a lower level when the simulated *NIT* encourages them to do so because the restrictions on the nature of job offers are beyond the control of any single household or small group of households. Only when a truly general desire for shorter hours arises will employers have sufficient incentive to change their offers. An experimental study will produce no such general desire, but only a desire on the part of the households in the sample.

If this is correct, then an experiment of the type discussed will not provide an accurate prediction of the shift in the supply curve. What such an experiment will provide is information about the responses of a small number of households to the *NIT*, given that the institutional elements in the labor market are beyond their control. These institutional elements are *data* for an experimental study, but they are *variables* for a national policy. Thus there is no reason to expect an equivalence in the responses of households in these two situations.

The assertion that households will respond to an experiment in the same way as they would to a national policy (which seems to be implicit in the article by Orcutt and Orcutt) reflects a misunderstanding of the nature of the choices open to households. The income-leisure apparatus traditionally employed in analyzing work effort responses is only useful in examining general changes in preferred levels of work effort, such as would result from a *NIT* policy of wide

coverage. It is not very useful in examining how a single household (or a small number of households) will respond to a policy when it is the only one subjected to this policy. The opportunities open to a household are simply not the same in these two situations.⁹ In the former case, it is legitimate to proceed with the analysis *as if* each household could freely choose its amount of work effort, for employers are given incentive to adjust their offers to the preferences of workers. In the latter case, this is not a legitimate step because the members of a sample are restricted in their choices by the unchanged desires of all the other workers in the market.

In reality, there is, of course, some scope for individual variation in work effort. The wider the range of choices open to *individuals*, the more likely it is that an experiment would provide a clue to the actual shift in the supply curve which a policy of national coverage would produce. I have no idea how much individual variation would be possible for the participants in an experimental study. The only point I wish to make is that the variation possible when a national policy is implemented is surely greater than that possible for a participant in such a study. This means that an experiment cannot generate an unbiased estimate of the shift in supply which would be produced by a national policy.

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⁹ The important distinction between the situation from the viewpoint of the individual and from the viewpoint of the group is concisely pointed out by Milton Friedman: "The important point is that the individual is like the perfect competitor: to each individual separately, the number of hours of work per week may be fixed, yet the level at which it is fixed is the result of the choices of the individuals as a group" (p. 205).